

these clients to propose transactions. The MSP makes use of a certificate authority, which is a pluggable interface that verifies and revokes user certificates upon confirmed identity. The default interface used for the MSP is the Fabric-CA API; however, organisations can implement an external certificate authority of their choice. This is another feature of Hyperledger Fabric that is modular.

Hyperledger Fabric supports many credential architectures, which allow for many types of external certificate authority interfaces to be used. As a result, a single Hyperledger Fabric network can be controlled by multiple MSPs, where each organisation brings its favourite.

What the MSP does: To start, users are authenticated using a certificate authority, which identifies the application, peer, endorser and orderer identities, and verifies these credentials. A signature is generated through the use of a signing algorithm and a signature verification algorithm. Specifically, generating a signature starts with a signing algorithm, which uses the credentials of the entities associated with their respective identities, and outputs an endorsement. A signature is then generated, which is a byte array that is bound to a specific identity.

Next, the signature verification algorithm takes the identity, endorsement and signature as inputs and outputs 'Accept' if the signature byte array corresponds with a valid signature for the inputted endorsement, or outputs 'Reject' if it does not. If the output is 'Accept', the user can view

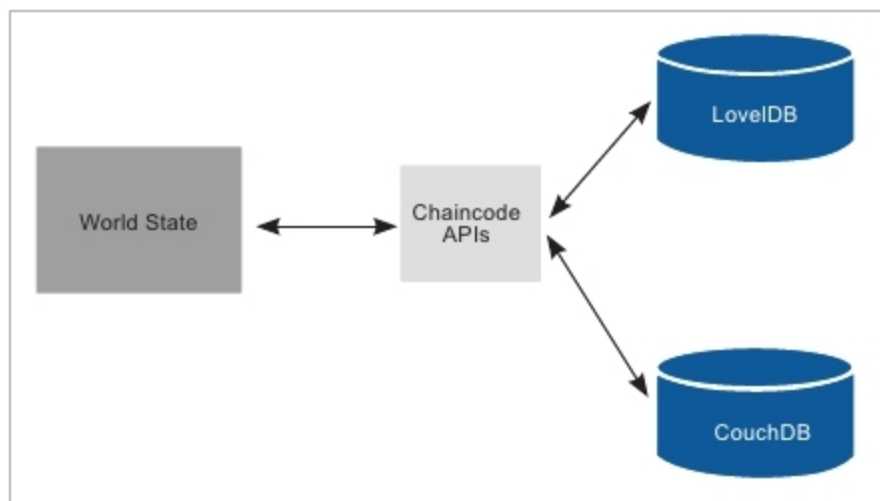


Figure 2: State database

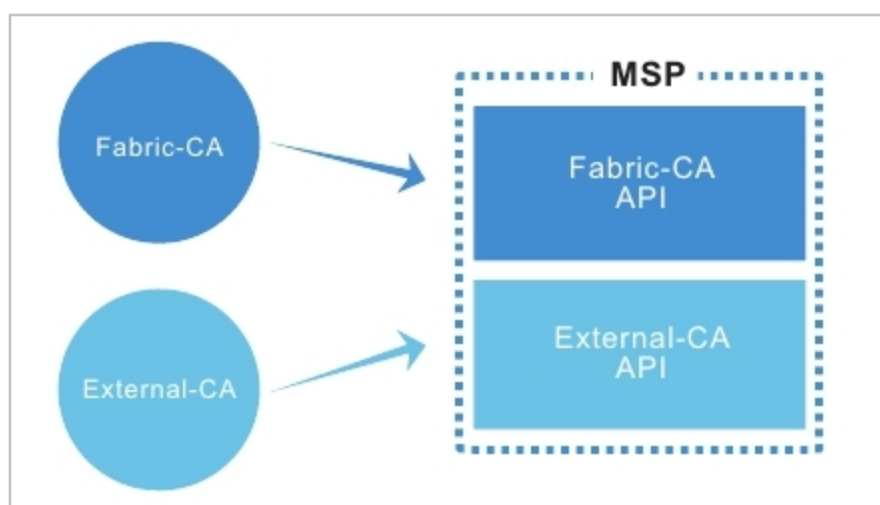


Figure 3: Membership service provider (MSP)

the transactions in the network and perform transactions with other actors in the network. If the output is 'Reject', it implies that the user has not been properly authenticated and is not able to submit transactions to the network, or view any previous transactions.

Fabric-Certificate Authority (Fabric-CA)

In general, certificate authorities manage enrolment certificates for a permissioned blockchain. Fabric-CA is the default certificate authority for Hyperledger Fabric and handles the registration of user identities. The Fabric-CA is in charge of issuing and revoking enrolment certificates (E-Certs). The current implementation of Fabric-CA only issues E-Certs, which supply long term identity certificates. E-Certs issued by the enrolment certificate authority (E-CA) assign peers their identity, and give them permission to join the network and submit transactions.

Getting involved with the Hyperledger Fabric project

Hyperledger Fabric is an open source project, where ideas and code can be publicly discussed, created and reviewed. There are many ways to join the Hyperledger Fabric community. You can join the weekly meeting on Fabric Documentation or other Hyperledger Fabric-related get-togethers. The Hyperledger Community Meetings Calendar is a great resource to learn about the timings for these meetings. You can also join the Hyperledger Fabric mailing lists for technical discussions and announcements. Gerrit is used for submitting PRs and managing code reviews and check-ins. You can join the live conversations on Rocket.Chat (which is an alternative to Slack), using your Linux Foundation ID.

What's next?

In subsequent articles, we will look at Hyperledger Fabric with respect to transaction flow, installation and chaincode development. **END** 🐧

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